

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Machine Learning, Neuroinformatics, Foundations of Data Science, Advanced Machine Learning, Convex Optimization, Reinforcement Learning

PROJECTS

LFP2Vec audit | Python, PyTorch, Hugging Face transformers, wav2vec2, scikit-learn github.com/happyc0der/lfp2vec-audit

- Reproduced LFP2Vec (NeurIPS 2025) fine-tuning on public IBL and Allen Neuropixels data on a laptop: 0.74 ± 0.07 balanced accuracy vs 0.68 published.
- Traced the cross-lab collapse to a preprocessing mismatch (spectra diverge 768x above 300 Hz); a 100 Hz low-pass restores the published margin.
- Measured calibration under lab shift (ECE 0.10 in lab, 0.56 across, 0.35 after the fix), added an electrode-position control; 337 tests in CI.

Gemma 4 SWE agent (in progress) | Python, Google ADK, vLLM, LiteLLM, github.com/happyc0der/gemma-swe-agent
Docker

- Kaggle Gemma 4 Developer Agent (solo, in progress): declarative Google ADK agent that fixes GitHub issues with Gemma 4 31B; final ranking pending.
- swelite, a re-implementation of the organizers' unreleased evaluation harness (Docker sandboxes, 9 tools, ADK YAML compiler): 109 of 129 public tasks pass gold checks.
- Ran a 16 GB laptop GPU over SSH as a vLLM proxy; found ADK never compacts context mid-task, and temperature 0.7 cut identical retries from 13.5 to 5.1.

Biohub cell tracking (in progress) | Python, PyTorch, NumPy, SciPy, polars 2026

Builds on Igor Zharov's public notebook (Apache-2.0); the harness, classifiers and experiments are mine.

- Kaggle Biohub Cell Tracking: public leaderboard 0.952 against 0.947 for the public notebook it builds on, on free Kaggle GPU; final ranking pending.
- Offline harness runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos, matching the notebook to $+0.00011$.
- 3D CNN trained on 134 annotated divisions prunes false forks: predicted $+0.0047$ on a pre-registered holdout, measured $+0.005$ on the leaderboard twice.

Blockwave | Python, numpy, pygame-ce, PyTorch, PPO github.com/happyc0der/blockwave

- Falling-block game in Python and numpy with no image, font or audio files: SRS rotation, 7-bag, T-spins, 240 Hz fixed timestep, 300k engine steps/s headless.
- PPO agent on an empowerment reward that never sees score, lines or level, enforced by a static test; the pixel version sees an 88x88 crop.
- 465 tests in 40 s; after two identical 150M-step runs differed by 42%, withdrew every single-seed comparison from the README.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465. Final ranking pending, competition ends 2026-09-30.
- **Gemma 4 Developer Agent, Kaggle (Google DeepMind)**, 2026. In progress, ends 2026-12-02.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, vLLM, Ollama, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale